VIRTUAL 2315 CARTRIDGE FACILITY

Build and Install

V2315CF Build and Install

Overview

Building the Virtual 2315 Cartridge Facillity (V2315CF) involves building and/or modifying several components.

- The disk drive in the IBM 1130 is modified to stop the heads from loading down onto the spinning cartridge and to route signals for the cartridge unlock switch and Unlock lamp

- A version of George Wiley's RK-05 Emulator is constructed and modified with a different front panel and change to the microSD connection. New code is loaded onto the FPGA and Pi Pico. This becomes the V2315CF main unit.

- Options *A2 RK-05 to RK-05 Adapter*, *T1 M930GW Terminator Board* and *A7 RK-05 Emulator Power Supply* are built

- Build the Power Distribution Unit involving a custom PCB and components that connect together the parts of the power system.

- Buy 12V motorcycle battery

- Buy a 12V timer module and configure it for 40 second ride-through after primary power drops

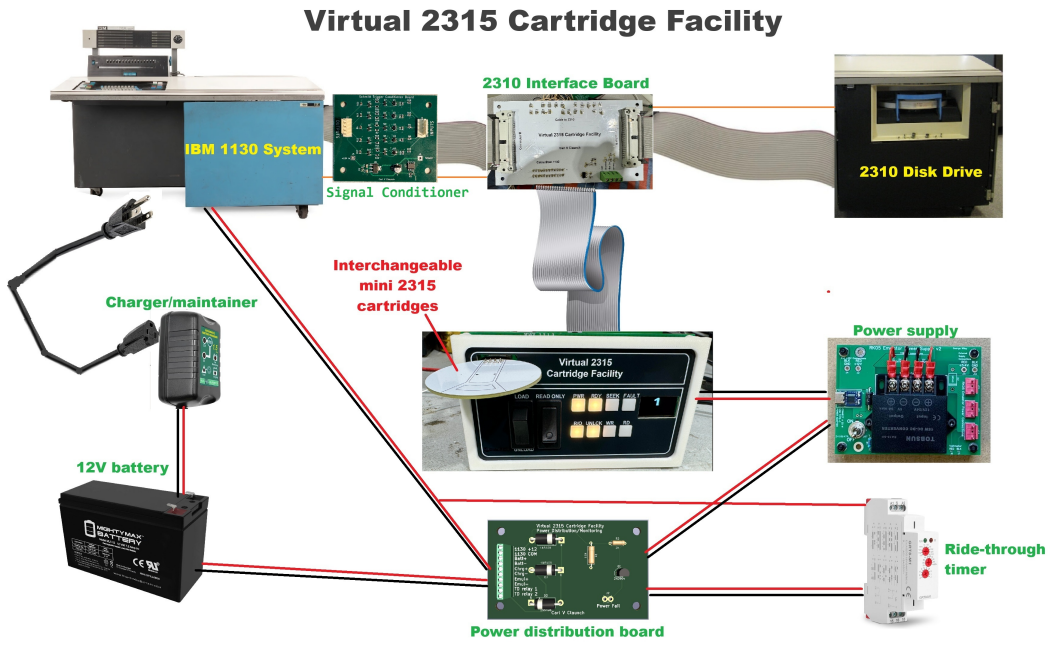
- Build the 2310 Interface Board, involving a custom PCB and components. This involves creating SLT male and female connectors.

V2315CF Build and Install

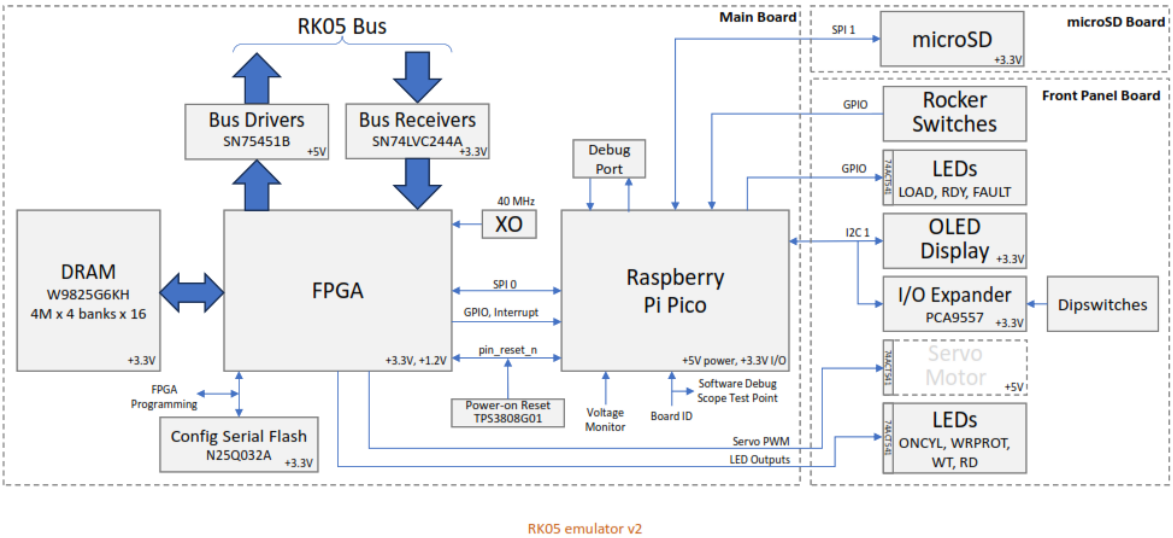
- Install the 2310 Interface Board in the 1130, connect signal cable from 1130 and attach new cable to disk drive. Wire up unlock lamp and cartridge handle unlock from drive.
- Place 12V battery and trickle charger inside 1130.
- Build housing to place V2315CF main unit on top of the internal disk drive, underneath the top cover of the 1130.
- Modify the T1 M903GW Terminator Board
- Run all wiring between units
- 3D print parts for mini 2315 cartridge holders
- Build boards for mini 2315 cartridges, involving custom PCB and components
- Install python programs on a personal computer
- Build adapter to mount mini-cartridge on PC over USB
- Load some microSD cards with disk images

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Block Diagram of the V2315CF



The central box in the diagram is the main unit of the V2315CF. Mini 2315 cartridges are plugged into the slot on the upper left.



V2315CF Build and Install

The 2310 Interface Board routes signals between the IBM 1130 System, the V2315CF main box and the 1130's internal disk drive. It also contains the Real/Virtual mode switch and circuitry to monitor and control the Unlock lamp on the IBM 1130 console.

A Signal Conditioner board is mounted to the side of the 1130 logic gate A to intercept some signals coming on the cable from the V2315CF in order to clean up the quality of the signals before they pass into the Solid Logic Technogy (SLT) circuitry.

The disk drive grounds a wire to the V2315CF when the drive has unlocked the handle allowing someone to insert/remove 2315 cartridges. The wire is connected to the 2310 Interface Board and another wire from that board is connected to the lamp on the 1130 console. The lamp is controlled either by the disk drive or V2315CF.

The power system of the V2315CF has a battery which is intelligently trickle charged whenever the IBM 1130 System is physically connected to building power, even if the 1130 system is switched off. The IBM 1130 system, when powered on, produces 12V power which is connected to the power supply of the V2315CF.

The 1130 supply normally powers the main unit and the 2310 Interface Board. The battery and 1130 system power are both connected to a Power Distribution Board, which in turn passes the power to the 2310 Interface Board and V2315CF main unit.

V2315CF Build and Install

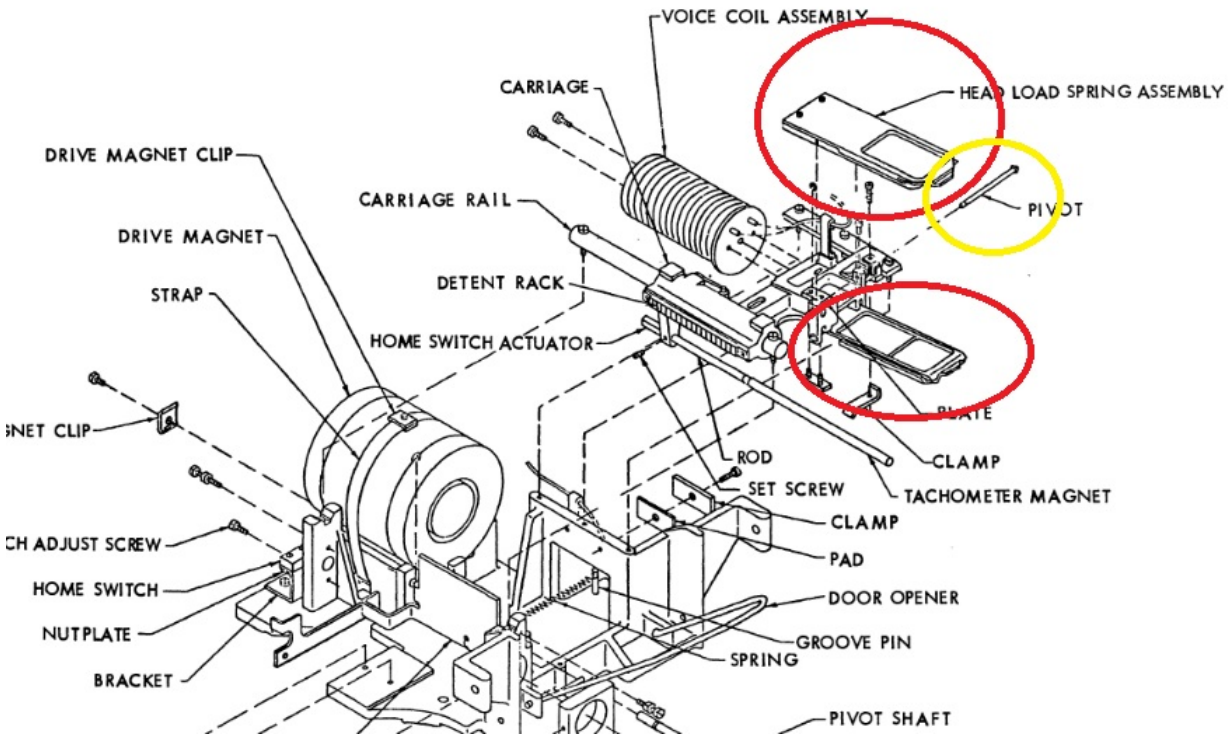
When the IBM 1130 system power fails, it is detected and a signal sent to the V2315CF which does an unload of the disk image back to the Mini 2315 Cartridge. A timer keeps the battery connected to the main unit for about 45 seconds, long enough for the unload, then drops power until the 1130 system power returns.

If the mini cartridge was set to Read Only mode and power fails, it does NOT write the data back to the cartridge. It stays consistent with the intent of the Read Only state.

V2315CF Build and Install

IBM 1130 disk drive modification

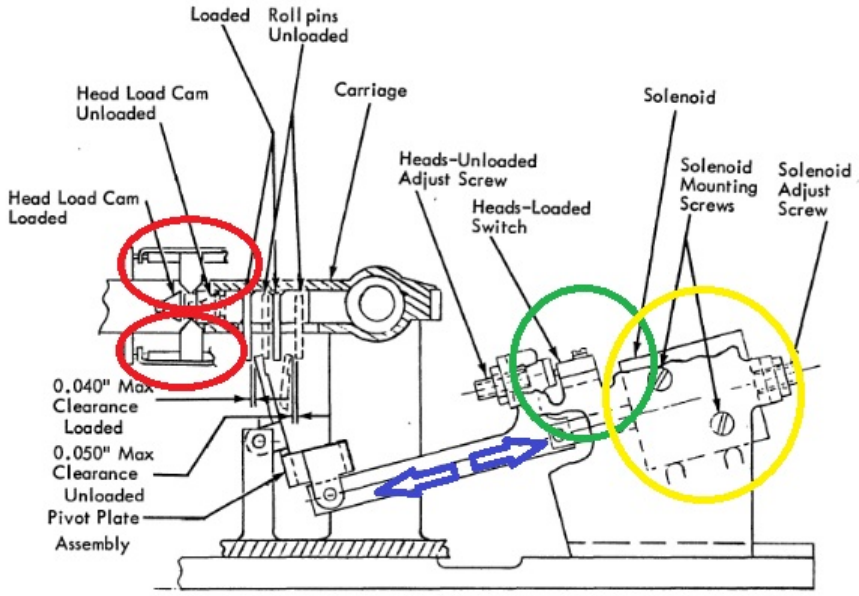
The drive must be modified so that the heads do not push down to load onto the spinning platter. This involves the removal of two head load spring assemblies and their pivot pin from the drive.



Without those head load spring assemblies, the heads stay far above the disk platter surface, held up by spring force on the holder for the heads. The assemblies pivot inward to force the head holders to move onto the platter, where they fly about 70 microinches above the oxide coating on a cushion of air.

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The diagram below shows the mechanism that activates the head load spring assemblies to pivot.



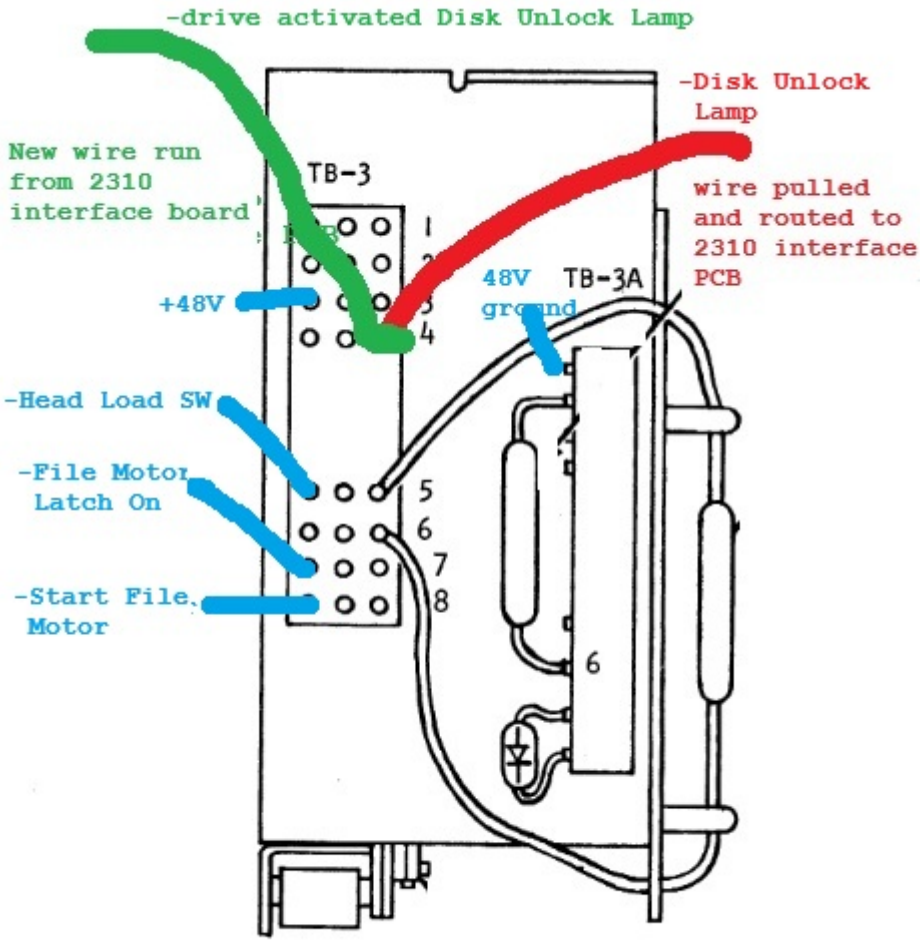
A solenoid (yellow circle) pulls on a rod that moves back and forth (blue arrows), pivoting to move a plunger left and right which pushes up on roll pins (red circles) on the rear of the head load spring assemblies, causing them to pivot. A microswitch (green circle) detects that the solenoid has pulled in and is forcing the heads to load.

The disk drive activates a wire when the cartridge loading handle is unlocked, which causes the DISK UNLOCK lamp on the console of the IBM 1130 to the left of the keyboard to illuminate. That signal is also detected by the V2315CF as it logically loads and unloads virtual cartridge images.

V2315CF Build and Install

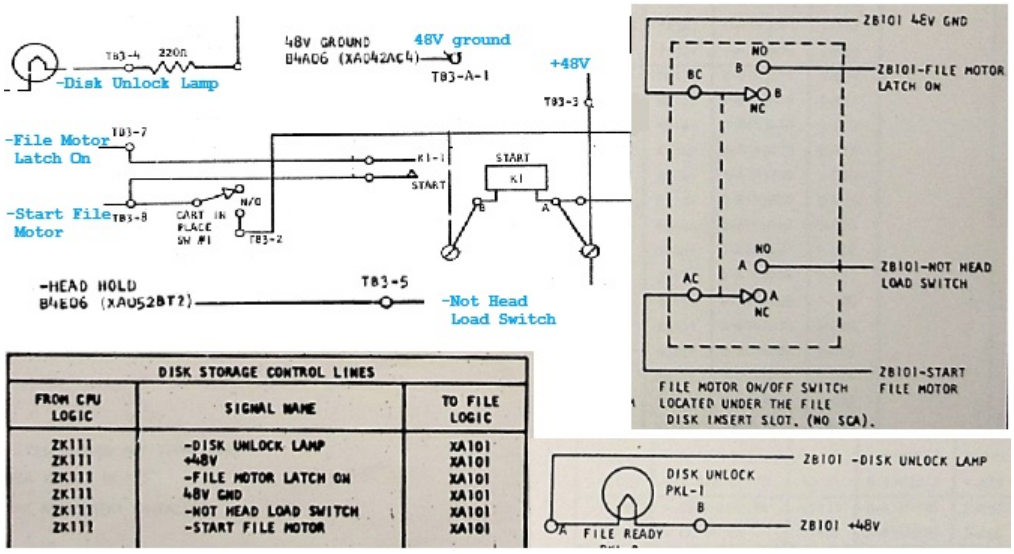
We have to modify wiring at the disk drive so that we can light the DISK UNLOCK lamp when in virtual drive mode as well as pass along the signal to light the lamp when in real mode. The lamp is powered by +48V and the signal is pulled to ground to energize the lamp.

We pull the wire that comes from the DISK UNLOCK lamp into TB3 and extend it so that it can be connected to the 2310 Interface PCB at connector J4 terminal 3. The new wire is run between TB3-4 and connector J4 terminal 2 on the 2310 Interface board.



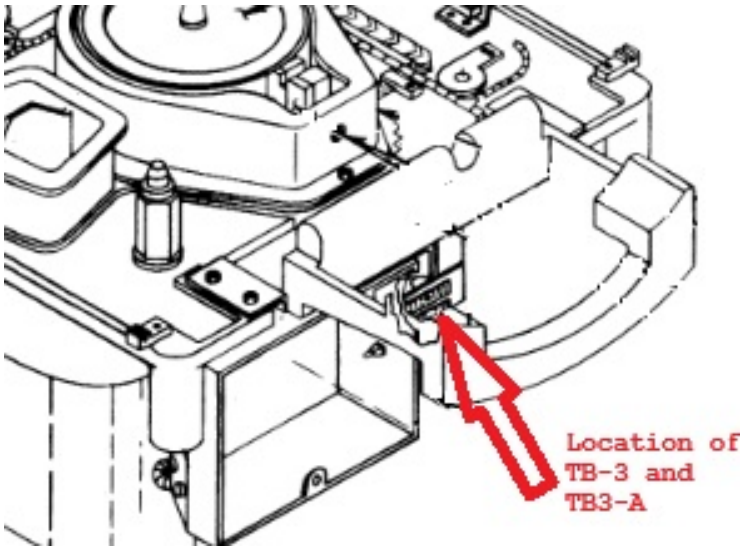
V2315CF Build and Install

The disk wiring into the IBM 1130 is very confusing in the IBM documentation. The diagrams below show the reconciliation with the same names for signals on all charts.



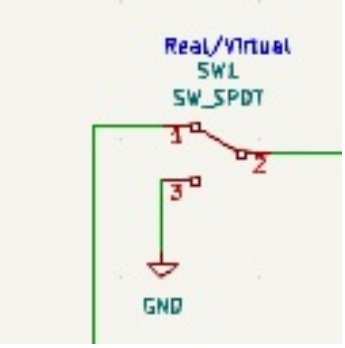
DISK STORAGE CONTROL LINES		
FROM CPU LOGIC	SIGNAL NAME	TO FILE LOGIC
ZK111	-DISK UNLOCK LAMP	XA101
ZK111	+48V	XA101
ZK111	-FILE MOTOR LATCH ON	XA101
ZK111	48V GND	XA101
ZK111	-NOT HEAD LOAD SWITCH	XA101
ZK111	-START FILE MOTOR	XA101

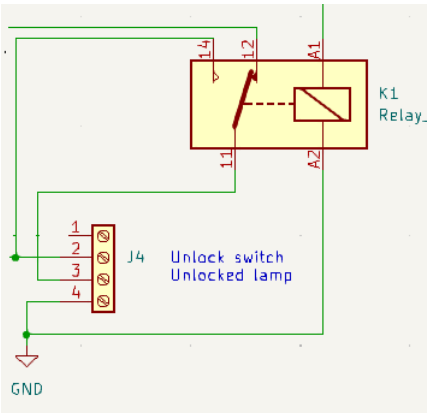
TB3 is located in the front of the disk drive as shown below:



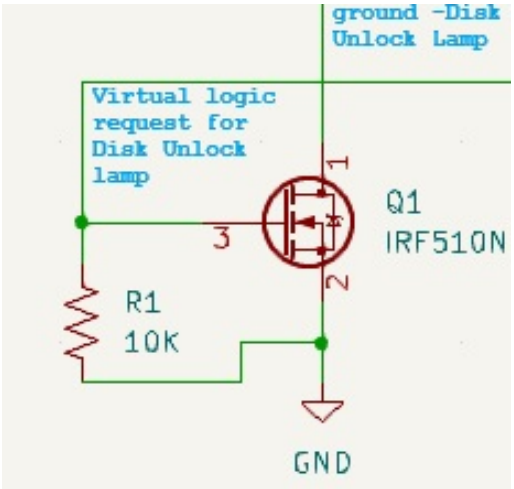
V2315CF Build and Install

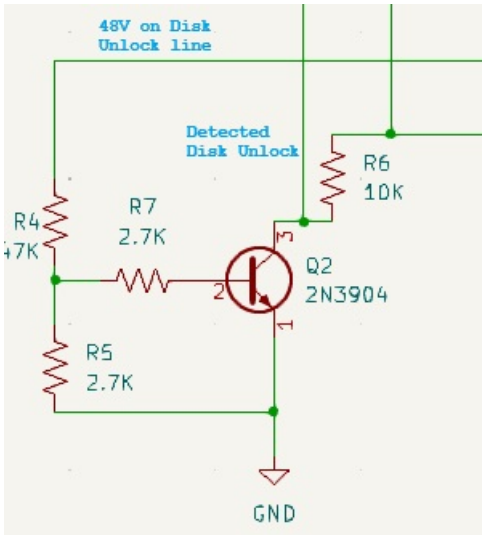
The 2310 Interface PCB has circuitry on it to pull the wire to the DISK UNLOCK lamp to ground when we want it lit. This includes a MOSFET for the drive when in virtual mode, and a relay to connect the - Disk Unlock wire from the disk drive to the DISK UNLOCK lamp when in real mode. It also has a transistor sensing circuit to detect when the drive has activated the lamp signal for use in the V2315CF logic.





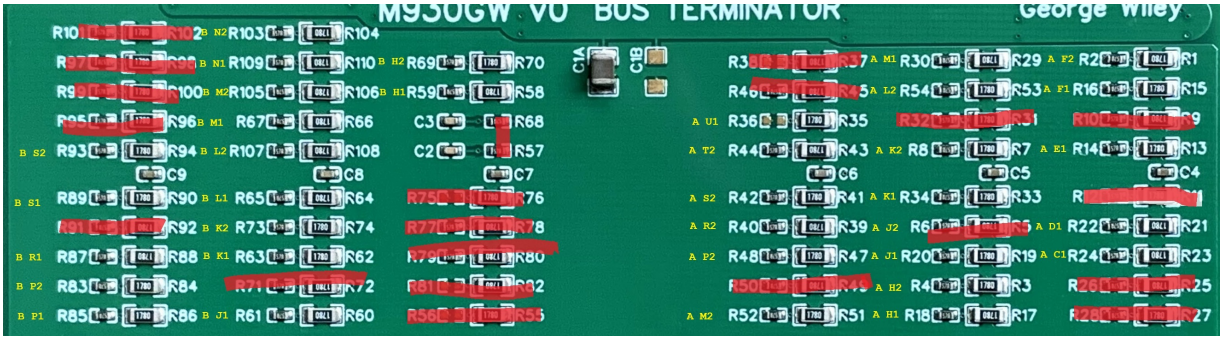
The mode switch activates the relay, either connecting the drives -Disk Unlock wire to the console DISK UNLOCK lamp, or connecting the lamp to the MOSFET driver. The transistor senses whether the -Disk Unlock line from the drive is active.





V2315CF Build and Install

T1 M903GW Terminator Board modification



The board as deivered from George Wily has 178 and 383 ohm resistors installed in most signal positions. There are a few that don't have a resistor installed because of their special use in the RK-05 emulator.

Remove all the resistors from the board, leaving the capacitors. Clean up the pads.

You will need 34 pairs of resistors to install on the board. The positions that have a red line across don't get any resistors because those signal pins are not used with the V2315CF.

Substitute all 1206 size resistors with 169 ohm components. Replace all 0603 size resistors with 249 ohm parts. This adjusts the impedance to match the ribbon cabling.

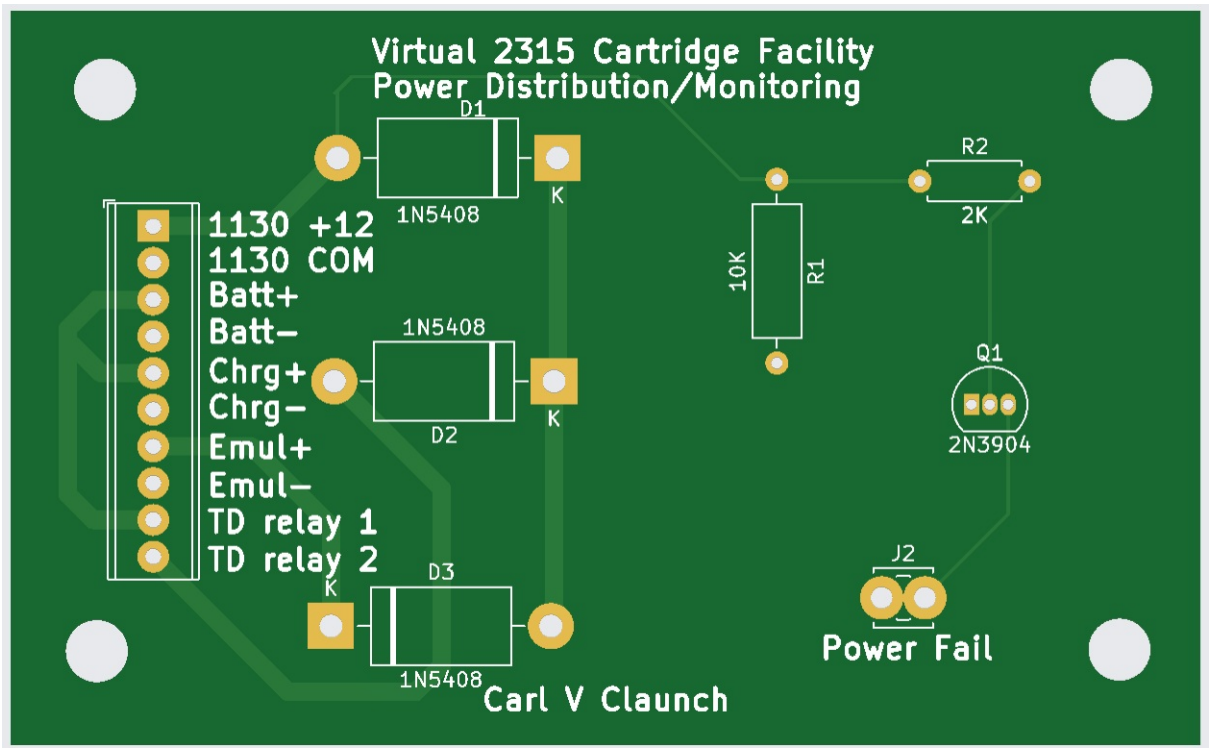
V2315CF Build and Install

Build Power Distribution Unit

The power distribution unit (PDU) interconnects the 1130 system power, intelligent trickle battery charger, 12V motorcycle battery, time delay relay module, and the power supply board for the V2315CF main unit.

It also detects when incoming power from the IBM 1130 has dropped, triggering an emergency unload of the disk contents from RAM back to the microSD card in the mini cartridge, informating the V2315CF main unit.

Get PCB. Install connectors, resistors, transistor and diodes.



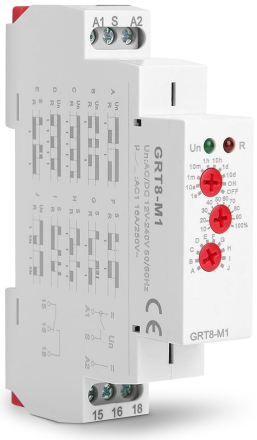
V2315CF Build and Install

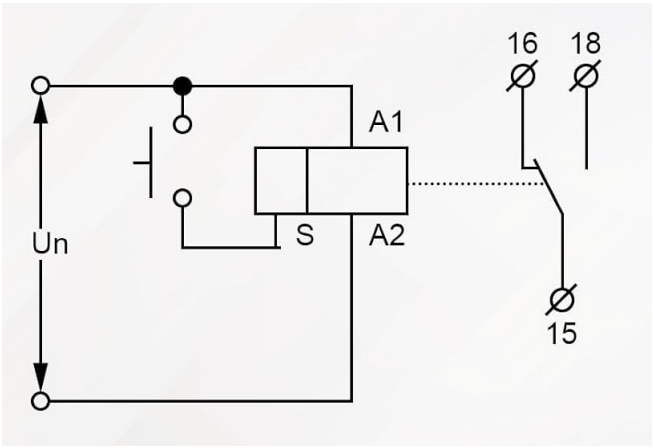
Configure 12V Timer Module

An off the shelf timer relay module is needed that has the ability to connect the 12V battery power to the PDU when it detects the +12V power from the IBM 1130 has gone off, then maintains that connection for a set time duration.

This powers the V2315CF main unit when the 1130 drops out long enough that it can read all the data currently in the RAM on the main unit and write it back to the virtual cartridge's microSD card.

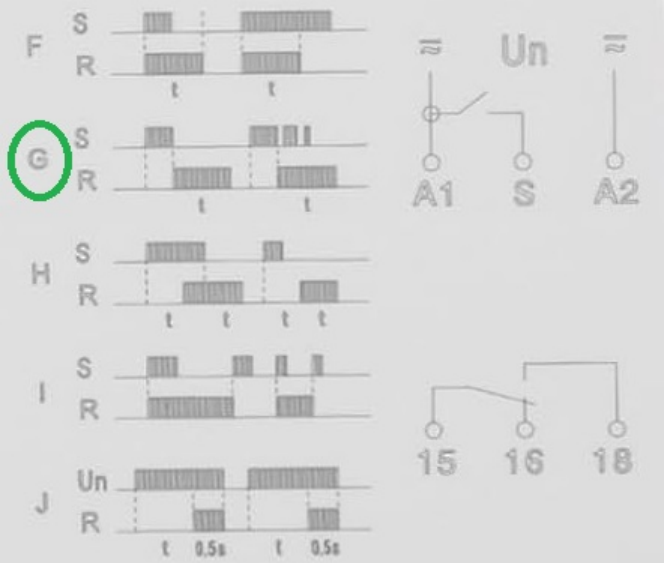
I purchased a suitable module on Amazon.com (GRT8-M1) but all we care about is switched contacts that can handle the 12V battery current, a trigger that detects the +12V from the 1130 and not the battery, plus a timer mode that activates the relay for 40 seconds.



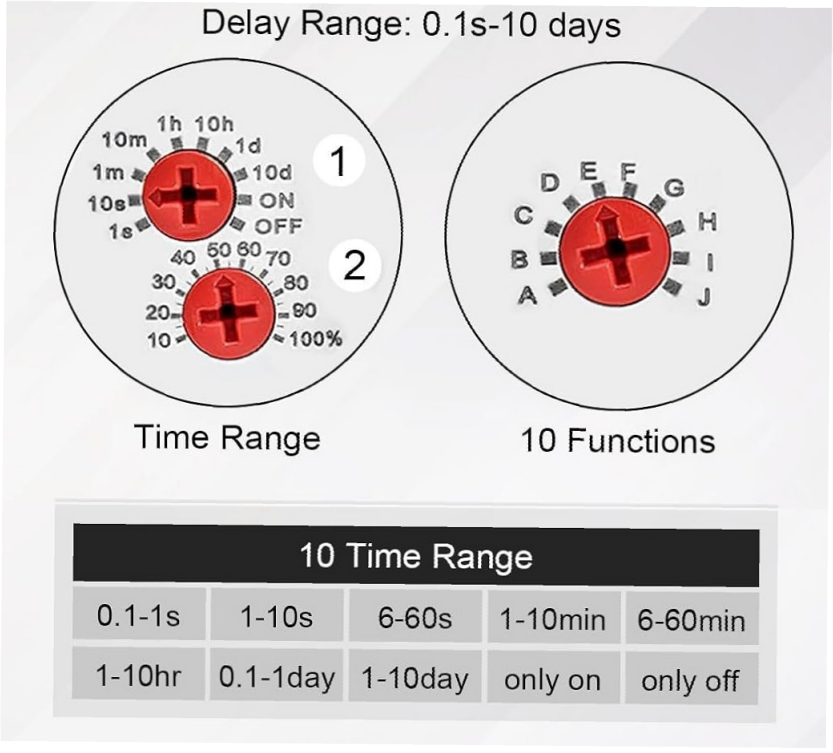


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On this unit, the timing function is selected on a rotary switch. We want choice G which turns on the relay for the set time when the power connected to the trigger terminal turns off.

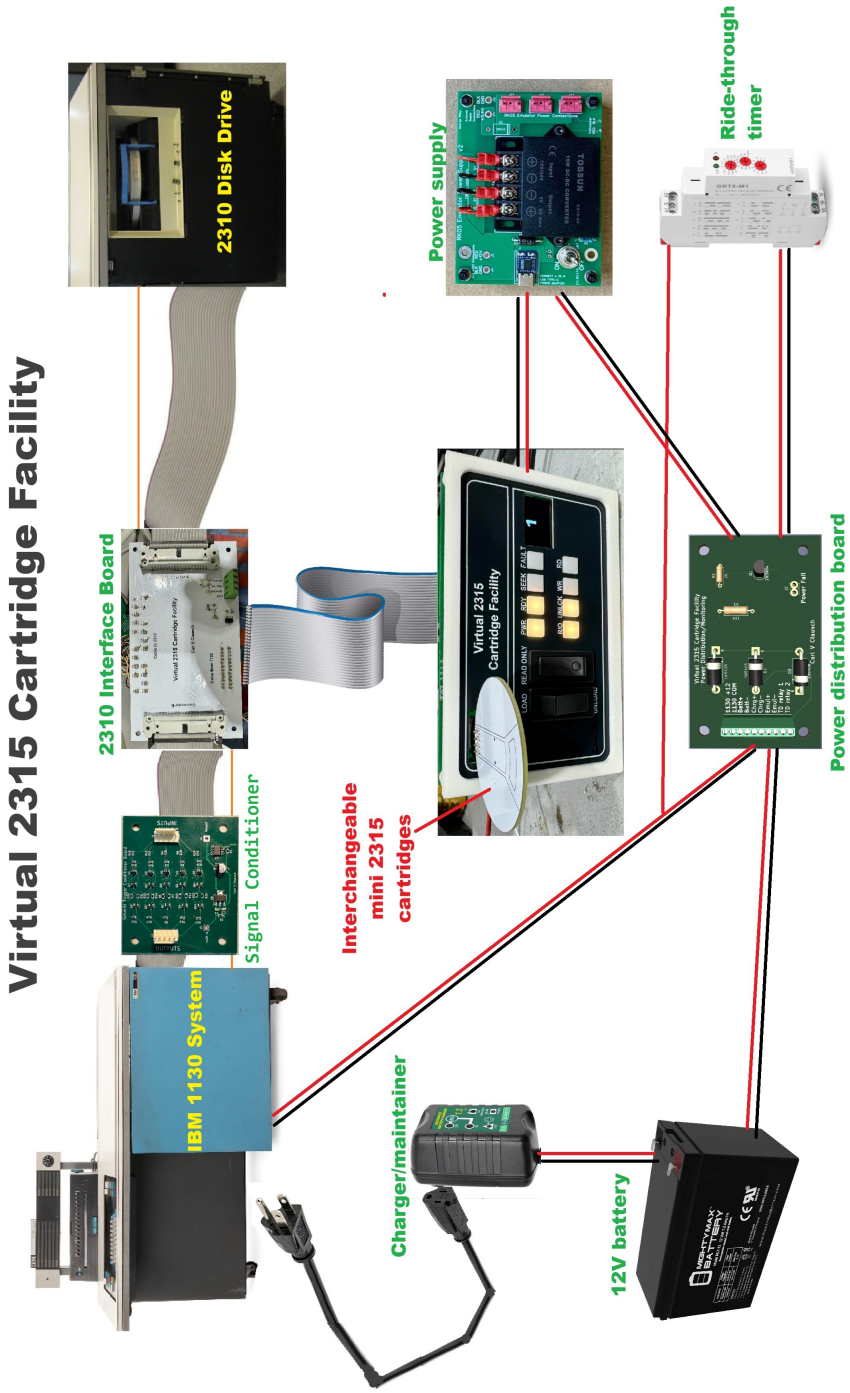


The time is set with a pair of rotary switches. I selected 40 seconds by turning the controls and timing it until it was right.



10 Time Range				
0.1-1s	1-10s	6-60s	1-10min	6-60min
1-10hr	0.1-1day	1-10day	only on	only off

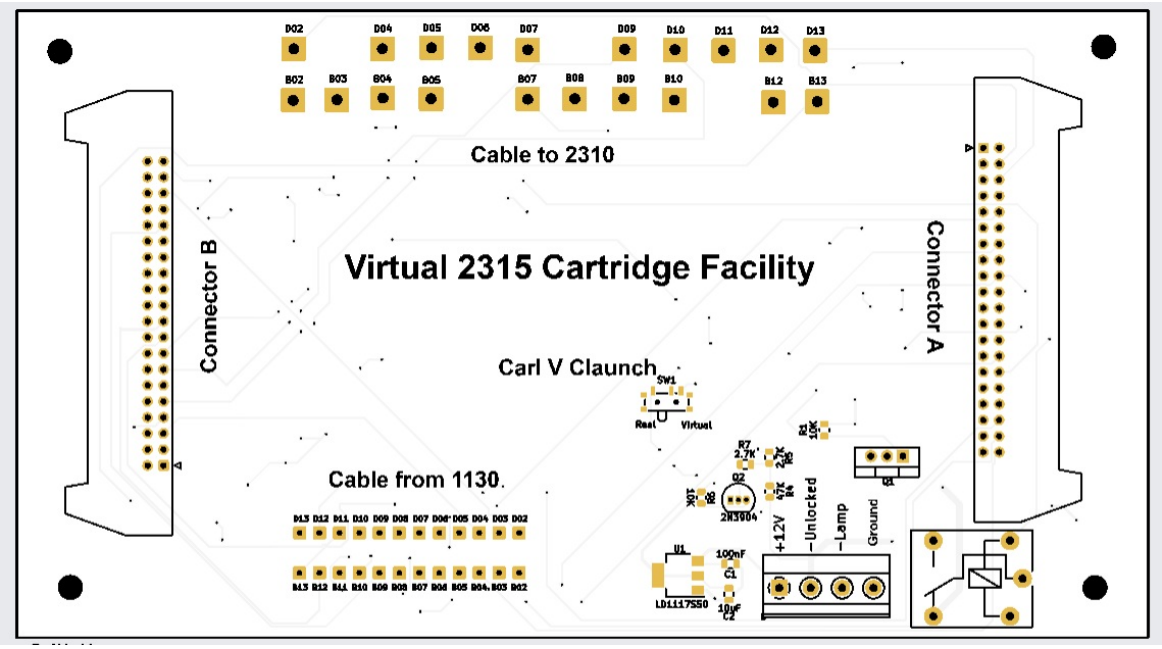
V2315CF Build and Install



V2315CF Build and Install

2310 Interface Board construction and installation

This PCB attaches to the two ribbon cables from the *A2 RK-05 to RK-05 Adapter* which was plugged into the V2315CF main unit. It also has a connector into which the signal cable from the IBM 1130 which originally plugs into the disk drive is instead plugged. At the top a set of wires are connected to an SLT connector that plugs into the internal disk drive in slot A2.



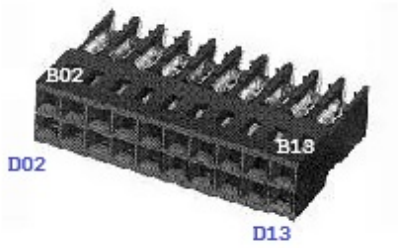
Install the transistors, resistors and voltage regulator onto the board. Solder the slide switch in place. Insert the relay and solder in place. Solder the J4 screw terminal strip in place. Solder down the two IDC-50 sockets.

V2315CF Build and Install

Build the female SLT socket and cable. Use 2- twisted wire pairs for the signal lines from the 2310 Interface Board pads across the top (Cable to 2310), connecting all the black wires together and soldering them to each other. Solder the white wires of each pair to pads D02, D04, D05, D06, D07, D09. D10, D11, D12, D13, B02, B03, B04, B05, B07, B08, B09. B10, B12 and B13.

The proper socket and pins to build an SLT socket are from TE Connectivity, using their MODIV 24 pin connector (1-102056-2) and 20 of the socket pins 26-22 AWG (104480-8). These can be bought from Mouser Electronics as items 571-1-102056-2 and 571-1044808.

An SLT socket, when viewed from the end, is organized in two rows of 12 pins. The rows are labeled D and B, the pin locations in a row run from 2 to 13.



Using the female socket as an alignment jig, insert male pins into the 2310 Interface PCB in the Cable from 1130 section near the bottom. The socket would be underneath the PCB and the male pins are pushed through to sit in the socket, then soldered in place. For the pins, use TE Connectivity CONN PC PIN SQUARE 0.025 GOLD (5-87623-4) which can be bought from Digikey, item number A100889CT-ND. These have a cross on the top so they won't push through the PCB.

V2315CF Build and Install

Drill holes for bolts to hold the PCB on the vertical rail inside the 1130 on the right side.

Pull the cable off the SLT card cage of the disk drive and extract the cable from the cooling plenum to free it up. Plug tht cable into the back of the 2310 Interface PCB, but make sure you have the orientation correct so that the signal from the 1130 that goes to D02 is plugging into the pin labeled D02 on our PCB.

You may have to use a continuity tester to be sure that the pin on the cable you think is D02 is connected to the 1130 gate A, compartment C1, card slot K6, pin D13.

Insert the newly constructed female connector and cable inside disk drive cooling plenum and plug into into card slot A02 of the card cage on the disk drive.

V2315CF Build and Install

Signal Conditioner board construction and installation

Install the operational amplifier, transistors, voltage reference chip, voltage regulator chip, resistors, capacitors and connectors on the printed circuit board.

Cut a rectangle of acrylic or other material as a base plate that the PCB will be mounted to. Drill four holes for the bolts and nuts that will attach the PCB to this mounting plate.

The two swing out frames inside the 1130 that holds all the SLT cards are called logic gates. Gate A is the one closest to the rear of the machine. They are both on a hinge on one side, so that they can be swung out to access them for servicing and also allows the two gates to be spread apart.

The side of logic gate A, at the hinge, is a good spot to place the PCB. Find a way to attach the mounting plate to the side of the logic gate and then install the PCB.

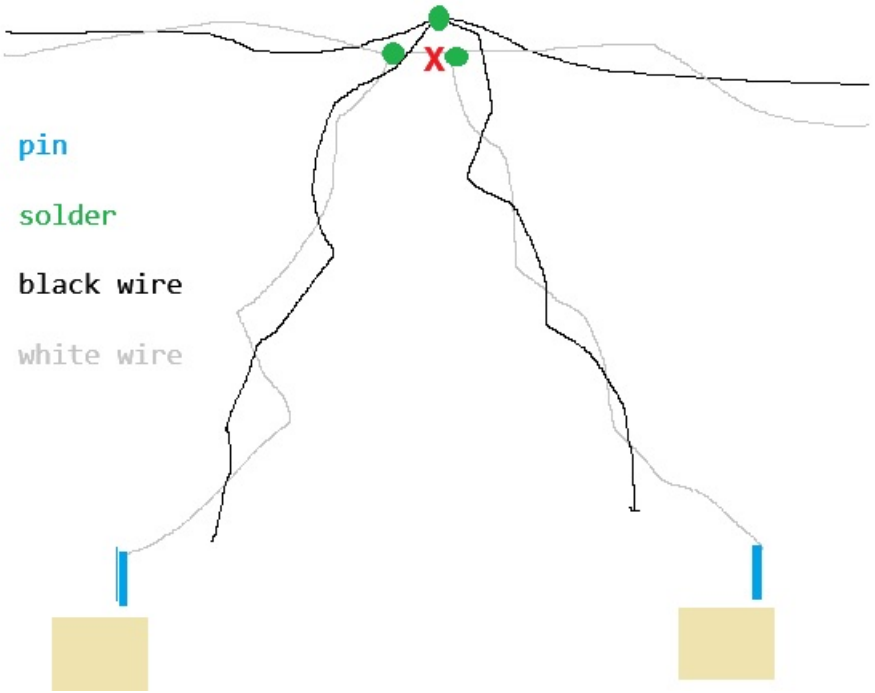
Run red and black stranded wires from the PCB (+12V and ground) and route them to the terminal strip inside the IBM 1130 where the power is distributed. Put a terminal connector on the end of each wire and then connect the red one to Terminal Block 6, on screws 8, 9 or 10. Connect the black wire to Terminal Block 5 screws 4, 5, 6, 7, 8, 9 or 10.

Remove the outer covering of the cable that runs from the V2315CF into the side of logic gate A, going into compartment C1 to deliver the signals to and from the disk. Do this near where the Signal Conditioner PCB is mounted.

V2315CF Build and Install

Spread the wire pairs out and locate the wire pairs for signals *-Sector Marker (D06), -Write Clock Phase B (B05)*, *-Read Clock (D02)*, *-Read Data (B03)*, and *-Access Ready (B07)*. Each of these five signals will be passed through one of the five channels of the Signal Conditioner.

Prepare the 5 pin connectors that hook to the input and output sides of the Signal Conditioner board. Using soare twisted wire pairs, attach the pin to a white wire and cut the black wire near the connector.

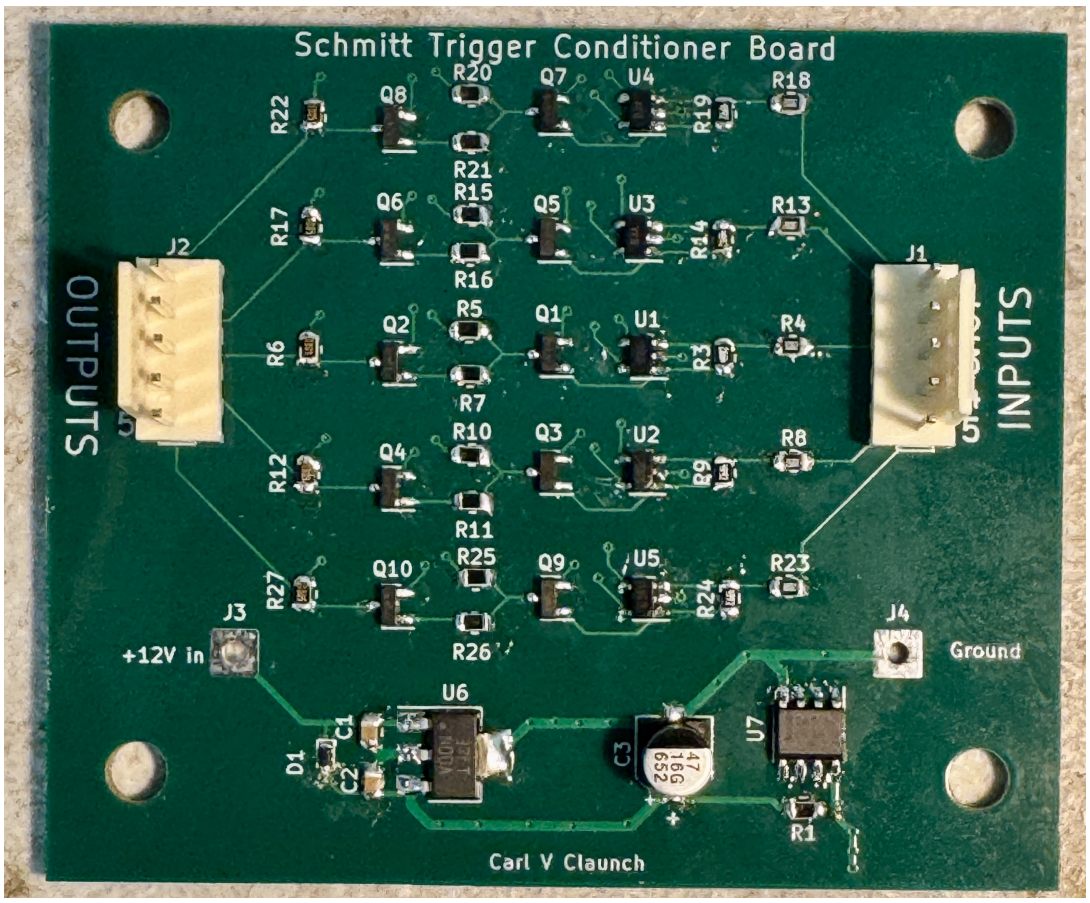


At the original cable where you located the five signal pairs, cut the white wire. Solder the white wire of the new pair for the input connector to the side of the white wire that is heading out to the 2310 Interface Board.

V2315CF Build and Install

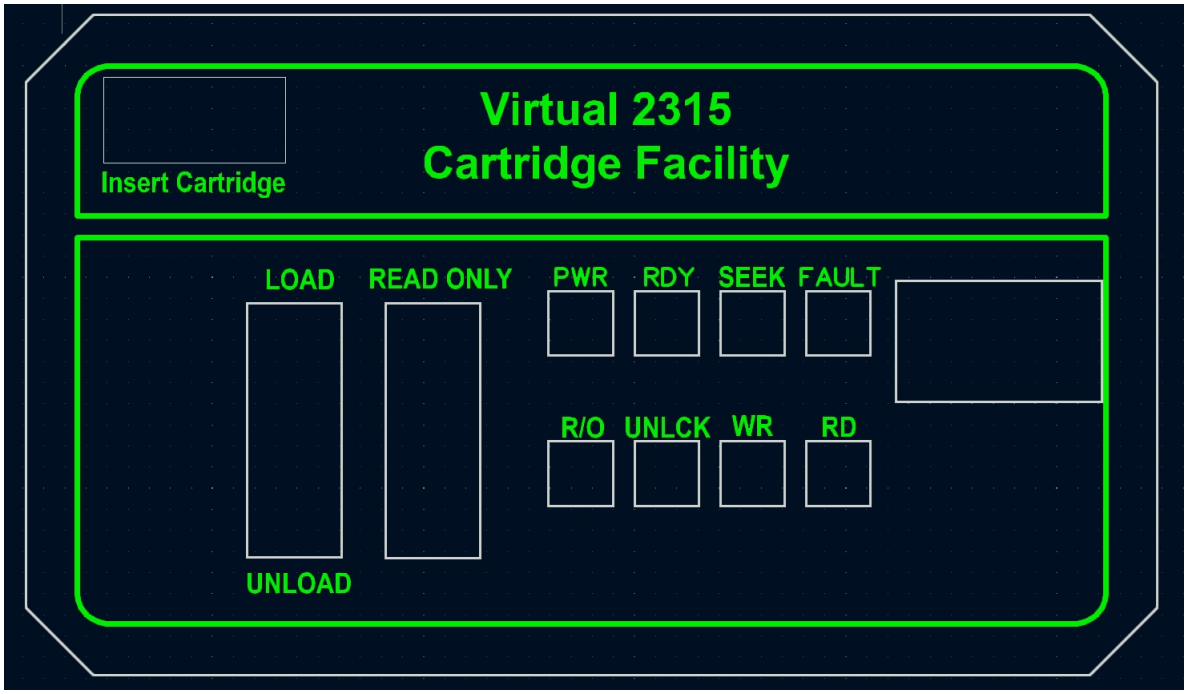
The white wire of the other new pair, going to the output connector, is soldered to the cut white wire side that proceeds into compartment C1 of logic gate A. Then the two black wires from the new pairs are soldered to the black wire of the original cable pair.

Do this for all five of the signals we are going to condition, making sure that the pin for the input and the output of each signal go to the same channel on the PCB.



V2315CF Build and Install

Modify RK-05 Emulator Kit to be V2315CF main unit



Replace front panel from kit with the V2315CF front panel.

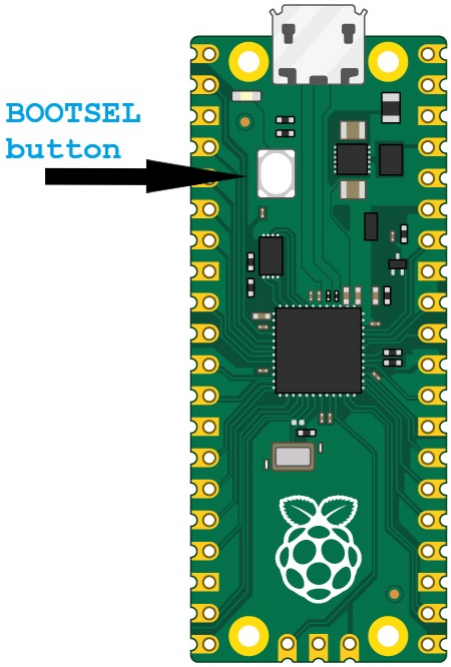
Instead of building the microSD PCB from kit, move the ribbon connector to face the opening in the front panel and secure in place.

Add wire connector JP27 inside the box which will connect to the power fail detection circuit on the power distribution unit (PDU) PCB, triggering the logic to do an emergency unload of the RAM to the virtual 2315 cartridge microSD card. This will be connected to connector J2 on the PDU.

V2315CF Build and Install

Load the file V2315_PICO.uf2 onto the Raspberry Pi PICO.

Pull Pico off the connector of the V2315CF main unit, hook up a USB cable to the PICO. While holding down the BOOTSEL button on the PICO, plug the other end of the USB cable into a computer.



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This will open the PICO as a disk drive on the computer. Copy the V2315_PICO.uf2 file onto that disk drive. When it has copied, unplug the USB cable from both sides.

Reinstall the PICO onto the board of the V2315CF main unit.

V2315CF Build and Install

To load the V2315CF logic into the FPGA, you will need a special cable and Diamond software from Lattice Semiconductor. The cable can be bought on eBay,, simply search for HW-USBN-2A and pick one that has a USB cable.

Diamond can be downloaded from latticesemi.com looking at their legacy FPGA software for Diamond. It is available for Linux and Windows computers. You also must request a free license to run the software from their website.

Attach the Lattice HW-USBN-2A to the V2315CF unit by plugging the ribbon cable into the connector marked FPGA programming. Plug in the power supply USB-C wall-wart that came with the RK-05 Emulator kit and use it with the *A7 RK-05 Emulator Power Supply* to power the V2315CF main unit.

You will download the file V2315CF_bitmap.bin by following the instructions in the RK-05 Emulator FPGA Programming V02 document included with this project.

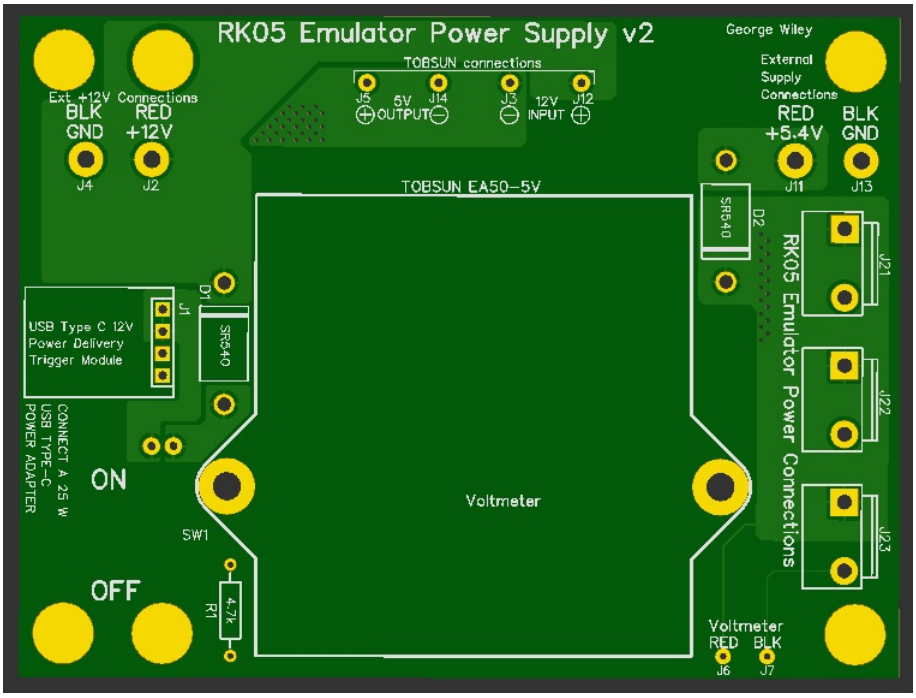
When the file is successfully downloaded to the storage for the FPGA initialization, power off the V2315CF, unhook the HW-USBN-2A cable, and shutdown the Diamond software.

The V2315CF main unit is now ready to connect and test.

V2315CF Build and Install

Build A7 RK-05 Emulator Power Supply unit

Install red wire from the top left +12V hole on the RK05 Emulator Power Supply board will be be connected to terminal screw 7 (Emul +) on the Power Distribution Board. Install a black wire from the top left GND hole on the RK05 Emulator Power Supply board to be connectd to terminal screw 8 of the Power Distribution Board.



Disconnect the On/Off toggle switch. It is onlyreconnected and a USB power brick is only used, with the red and black wires removed from the Power Distribution Board, when you are loading new firmware in the FPGA of the V2315CF.

Wiring up the entire system

Plug *A2 RK-05 to RK-05 Adapter* and the *T1 M930GW Terminator Board* into the rear of the V2315CF main unit.

Connect the A and B cables from the *A2 RK-05 to RK-05 Adapter* to the labeled connectors on the 2310 Interface PCB.

Connect the signal cable from the IBM 1130 into the rear of the 2310 Interface PCB. This cable would have originally been plugged into the internal disk drive SLT card cage in slot A2.

Connect the cable wired into the 2310 Interface board, route it through the cooling fan entry and plug it into slot A2 of the SLT card cage on the disk drive.

Connect the wire you extended from the DISK UNLOCK console light, where it had been removed from the disk drive terminal TB3, and connect it to connector J4 screw terminal 3 on the 2310 Interface board.

Connect the new wire you ran from the disk drive terminal TB-3 pin 4 to the 2310 Interface PCB connector J4 screw terminal 2.

V2315CF Build and Install

Run a red 18 gauge stranded wire from the 2310 Interface jack J4 terminal 1 to the main power distribution terminal block TB-6, putting a ring terminal on that end of the wire and connecting it to screws 8, 9 or 10, whichever is easiest mechanically.

Run black 16 gauge stranded wire from the 2310 Interface Board connector J4 terminal 4 to the main power distribution terminal block TB-5, putting a ring terminal on that end and connecting it to screws 4, 5, 6, 7, 8 or 9 depending on which is easiest.

Run a red 18 gauge stranded wire from terminal block TB-6, putting a ring terminal on that end of the wire and connecting it to screws 8, 9 or 10, whichever is easiest mechanically. Split the red wire into two, one of which has a ring terminal that is connected to the power distribution unit jack J1 terminal 1. The other wire is connected to the timer module terminal S.

Run an 18 ga stranded black wire to TB-5 with a ring terminal on one end which is connected to screws 4, 5, 6, 7, 8, or 9. The other end has a ring terminal that is connected to the power distribution unit PCB at jack J1 screw terminal 2.

Mount the 12V battery securely on the floor of the 1130 near where the 2310 Interface PCB will be mounted. Run red and black 18 ga stranded wires from the battery up to the power distribution unit. Split each wire into two wires. One red and black is connected to the PDU jack J1 screw terminals 3 and 4 respectively.The other red wire is connected to the timer terminal A1. The other black wire is connected to the timer terminal A2.

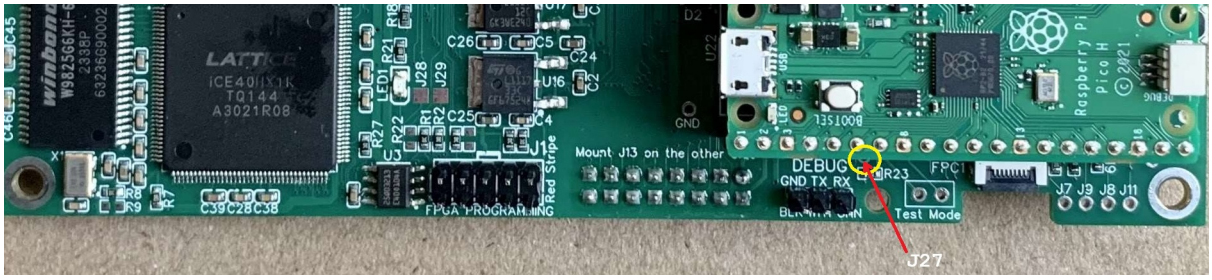
V2315CF Build and Install

Plug the intelligent battery charger into the convenience outlet on the right side of the IBM 1130 and connect its output wires to the power distribution unit jack J1 screw terminals 5 and 6 according to polarity.

Hook up the red wire from the *A7 RK-05 Emulator Power Supply* to the power distribution unit jack J1 screw terminal 7.

Hook up the black wire from the *A7 RK-05 Emulator Power Supply* GND input to power distribution unit jack J1 screw terminal 8.

Screw terminals 9 and 10 of the power distribution unit jack J1 are wired to the two switched relay contacts 15 and 18 on Timer module. Use ring terminals on the power distribution board end and appropriate connection to the timer relay module.



Run a wire from jack J2 of the Power Distribution Unit to V2315CF main box connector J27 (right pin of the pair labeled Test Mode) for power failure detection.

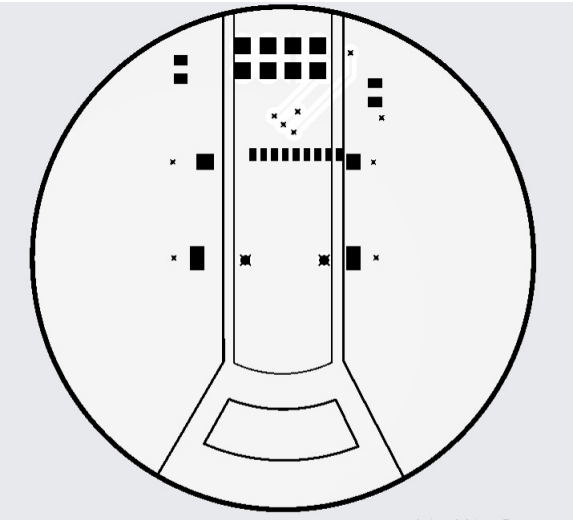
Run a red 18ga stranded wire from the Signal Conditioner board +12V connection to terminal TB-6 terminals 8. 9 or 10. Run a black stranded wire from the Signal Conditioner board ground connection to TB-5 terminals 4, 5, 6, 7, 8, 9 or 10.

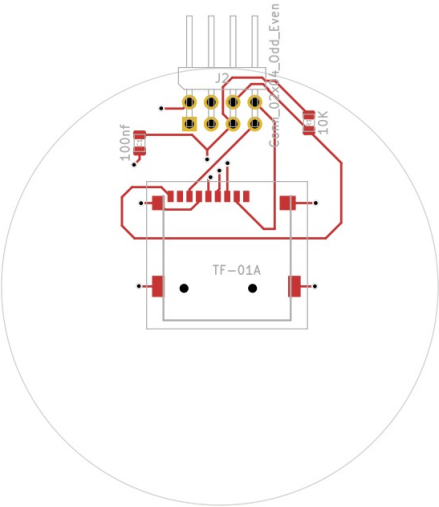
V2315CF Build and Install

Mini 2315 Cartridges as interchangeable file holders

These units plug into the connector on the front of the V2315CF main unit to provide the virtual disk cartridge image that will be used. A microSD card is installed inside the mini cartridge.

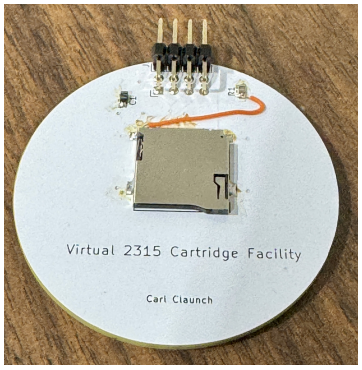
Order PCBs and install connectors, parts and microSD sockets. The backside of the PCB is designed to look like the top of a 2315 cartridge.





Insert these underneath the 3D printed mini 2315 cartridge objects, with the microSD card hidden inside the mini cartridge and only the bottom of the small PCB visible from beneath.





V2315CF Build and Install

CartInterface to read/write mini cartridges from PC

I provided some code in the project (CartInterface.uf2) that should be installed in another Raspberry Pi Pico which allows it to plug in a Mini 2315 Cartridge which appears as a disk drive to the PC when the USB cable from the Pico is plugged in.

A female 2x4 socket (2.54mm spacing) should be wired to the Pico. The Mini 2315 is plugged into the 2x4 socket then the Pico USB cable is plugged into the PC. The wiring of the socket to the Pico is:

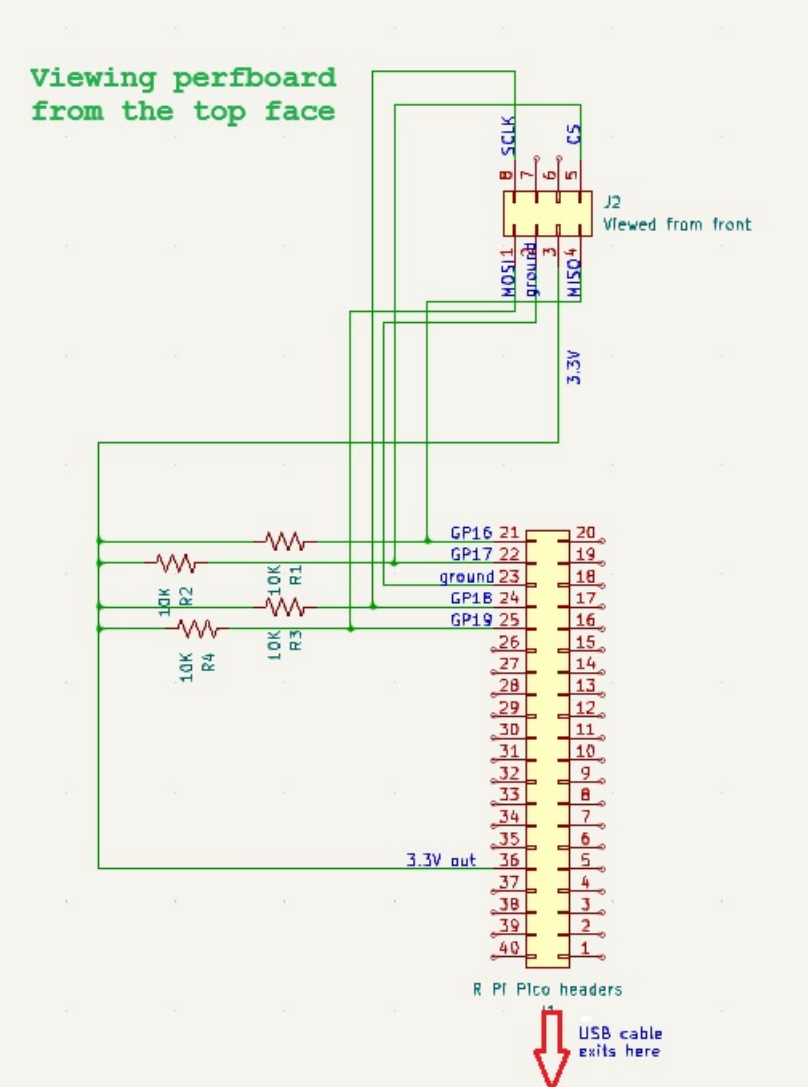
Pico Pin	2x4 socket pin	function	GPIO #
36	5	+3.3V	--
33	3 and 4	ground	--
14	2	clock	10
15	1	MOSI	11
16	7	MISO	12
17	8	select	13

The socket is oriented so that the 1 pin is down and left, when looking at the socket. Add 10K pullup resistors from pins 14, 15, 16 and 17 of the Pico to 3.3V (pin 36). The Mini 2315 cartridge is plugged in with the white face of the PCB up and the male header pins underneath the PCB. Put the PICO and socket in a small project box for convenient handling.

The CartInterface folder in the repository contains an Arduino sketch to load onto a Raspberry Pi PICO along with a schematic for how to build the tool. It allows you to plug a mini cartridge into the tool and plug the USB cable of the tool into a PC.

V2315CF Build and Install

The file system on the microSD card inside the mini cartridge is thus exposed to the PC as a disk drive from which you can read and write files. This is a convenient way to grab the disk image from a mini cartridge and work with it on the PC, as well as a handy way to install a 2315 disk image onto a mini cartridge for use with the V2315CF.



The file in the Mini 2315 cartridge has a name like xxxxx.dsk and it is in V2315CF format. Only one file with a .dsk suffix can exist on the microSD card at any time.

V2315CF Build and Install

Install support programs on a PC

I provided a number of programs that work with the V2315CF, which are included on the github repository for the project. Most are Python code, however they have been converted into .exe files for the PC in order to run them. The scripts are also provided in case one wants to use these on a non-Windows system.

convert2315.py - Opens a file (suffix .dsk) that is formatted for use with V2315CF and converts it to save as a new file (suffix .dsk) that is in the format used by IBM 1130 simulators.

convert1130.py - Opens a file (suffix .dsk) that is formatted for use with IBM 1130 simulators and converts it to save as a new file (suffix .dsk) that is in the format used by V2315CF.

listcartridges.py - Looks through a directory on the PC and reads every file with a .dsk suffix, listing the header information in the file if it is in V2315CF format.

showsector.py - Will open a disk cartridge file in V2315CF format and list the contents of a sector in hexadecimal.

CartInterface.uf2 - this file is loaded into a Raspberry Pi PICO to have it present a Mini 2315 cartridge as a disk drive on a PC when the PICO USB cable is plugged into the PC.

V2315CF Build and Install

Build enclosure and mount on 1130

Have acrylic laser cut to the included designs to build the enclosure for the V2315CF. Begin with the base plate onto which we will be mounting the power distribution unit PCB, the timer module, the trickle charger, and the *A7 RK-05 Emulator Power Supply*.

Place the two sides against the base plate and use the top, front and back pieces to help align then properly. Smaller rectangular rods of acrylic will form the braces. Use acrylic glue to fasten the sides to the base.

Place the front so that it sits on the edges of the sides and against the base plate. Glue it in place. Do the same with the back, leaving the top off to finish assembly.

WHAT ARE THE MOUNTS OR BRACKETS TO HOLD MAIN UNIT IN PLACE? DESCRIBE HOW TO MOUNT IT. COVER MODEL DISK DRIVE BUILD AND INSTALL>

Move the assembled enclosure, minus its top plate, to the 1130 and position it atop the cover of the internal disk drive. Make sure that all the cables are routed cleanly and that the top cover of the 1130 can close properly.

Attach the base plate to the 1130 so that the enclosure is firmly in place. Once all looks good, attach the top by drilling holes in rectangular rod segments that line up with holes you drill in the top piece. Glue the rod segments in place after ensuring that the holes line up. Glue nuts on the bottoms of the segment holes so

that bolts can thread into them through the top cover.

V2315CF Build and Install